

Identifying Turbulent/ Non Turbulent Interface In A Boundary Layer Data

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17th April, 2023

Abstract

The research delineates the Turbulent/Non-Turbulent Interface (TNTI) in boundary layer flows using Particle Image Velocimetry (PIV) data. A rigorous interrogation of mean velocity distributions against turbulent fluctuations unravels a TNTI. Analysis of Turbulent Kinetic Energy uncovers an intensification of turbulence downstream and highlights spanwise inhomogeneities. Turbulent Intensity metrics, calculated using both localized and free-stream velocities, quantitatively reflect the predominance of turbulence throughout the flow field. The study innovatively applies thresholding to the Total Kinetic Energy to demarcate turbulent regions, identifying a critical fluctuation threshold between 15-25% of the free stream velocity as the boundary for turbulent activity. Vorticity thresholds underscore rotational fluid motion regions, correlating high vorticity with turbulent zones. This approach elucidates the underlying mechanics of boundary layer turbulence, offering technical insights vital for computational turbulence modelling.

Keywords: Turbulent Kinetic Energy, Turbulent Intensity, Total Kinetic Energy, Vorticity, Critical Fluctuation Thresholding

Author roles: Guide: Prof. Alakesh Chandra Mandal ; Student: Debanjan Manna

Acknowledgements

I am profoundly thankful to Prof. Alakesh for his unwavering support and guidance, steering me through every challenges. My sincere gratitude goes to Mr. R Anvesh for sharing his Master's thesis, which has been a crucial asset in grasping the intricate problem statement. I must also extend my thanks to Kshitij Kaithal for engaging in meaningful discussions that have often enriched my understanding.

1 Introduction & Motivation

In fluid dynamics, turbulent flows, whether occurring naturally or engineered, such as in jets, wakes, and mixing layers, typically feature a turbulent region surrounded by non-turbulent fluid. These regions are demarcated by a **turbulent/non-turbulent interface** (henceforth **TNTI**), which is a dynamic boundary where non-turbulent fluid becomes turbulent.¹

The study of TNTIs is pivotal for understanding the mechanisms of turbulence interaction with laminar flows. These interfaces are dynamic zones where turbulent and non-turbulent regions meet, characterized by a sharp gradient in turbulence properties such as velocity and vorticity. The interface's evolution is governed by complex processes that include entrainment², where non-turbulent fluid is drawn into the turbulent region, enhancing the turbulence propagation and affecting the flow stability.

Mathematically, *the TNTI can be described by the discontinuity in the turbulence intensity and its derivatives*

¹The TNTI significantly influences the turbulence kinetic energy budget by affecting the production, transport, and dissipation of energy within the flow.

²Entrainment describes the process of drawing or transporting ambient fluid into a flow region, such as from non-turbulent to turbulent areas, enhancing mixing and energy exchange.

across the boundary Pope (2000). This interface significantly influences the turbulent kinetic energy (TKE)³ budget by affecting the production, transport, and dissipation terms. The interface is not static but evolves through mechanisms that involve both molecular viscosity and larger-scale interactions, contributing to its highly contorted and fractal nature.

The precise characterization of these interfaces is crucial for accurate modeling in computational fluid dynamics (CFD), especially for simulations that aim to resolve complex flows in engineering and environmental applications. Advanced techniques such as Direct Numerical Simulation (DNS)⁴ are employed to capture the detailed dynamics at these interfaces, allowing for a better understanding of the multi-scale interactions that occur and providing insights into effective strategies for turbulence management and control.

Research on TNTIs not only enhances fundamental theoretical knowledge but also supports practical advancements in optimizing systems where turbulence plays a critical role, such as in reducing drag in aerospace engineering or improving the mixing efficiency in chemical reactors.

1.1 Literature Review

The TNTI serves as a crucial boundary in fluid dynamics, delineating the chaotic, eddying motions of turbulent flow from the smoother, orderly laminar flow. This interface is instrumental in studying the mechanisms of turbulence propagation and fluid entrainment, which are pivotal for understanding flow stability and turbulence enhancement in various fluid dynamics applications. Balamurugan, Rodda, Philip, and Mandal (2020) delve into the complexities of these interfaces within spatially evolving turbulent mixing layers, emphasizing their role in flow dynamics.

At the TNTI, there exists a pronounced discontinuity in vorticity (ω)⁵ and velocity, serving as indicators of intense dynamic activity. Corrsin and Kistler (1955) initially identified these sharp changes using experimental techniques, paving the way for a deeper understanding of the interface's physical characteristics. Subsequent studies utilizing Direct Numerical Simulation (DNS) have confirmed these findings, revealing that these discontinuities can indicate shear zones⁶ critical for the transition from laminar to turbulent flow states.

The interface significantly influences the turbulence kinetic energy budget by affecting the production, transport, and dissipation of energy within the flow. Pope (2000) provides a comprehensive analysis of these interactions, offering insights into the theoretical modeling of turbulence dynamics at these boundaries.

The accurate representation of TNTIs is crucial for developing turbulence models in Computational Fluid Dynamics (CFD). Insights from detailed studies of TNTIs contribute to improved model precision, especially in predicting phenomena like flow separation and reattachment in complex geometries Durbin and Reif (2011). This accuracy is critical for applications ranging from aerospace engineering to industrial process optimization, where control over turbulence is paramount.

Anvesh (2016) provides a thorough examination of the TNTI in a spatially evolving mixing layer. It leverages Particle Image Velocimetry (PIV) to collect the velocity data, which are then subjected to further processing. The research specifically addresses the entrainment mechanisms and their impact on turbulence intensity and propagation, offering substantial insights into the discontinuities of turbulence properties across TNTIs and their influence on the turbulence kinetic energy budget.

Similarly, Bisset, Hunt, and Rogers (1999) delves into the characterization of TNTIs in turbulent shear flows, with a focus on the outer boundaries of turbulent zones in far wakes. The findings, derived from comprehensive DNS data, enhance understanding of how these interfaces interact with surrounding non-turbulent fluids. This includes detailed analysis of vorticity surfaces and their role in the entrainment process, advancing knowledge on the intermittent and dynamic nature of turbulence at these critical boundaries.

1.2 Objective Of The Present Work

Our aim is to find the TNTIs in Boundary layer flow over a flat plate.

³TKE, k , is defined as $k = \frac{1}{2}(\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$, where u' , v' , and w' represent the velocity fluctuations in the x, y, and z directions, respectively.

⁴DNS resolves all the scales of motion in a turbulent flow without any modeling assumptions, directly solving the Navier-Stokes equations.

⁵Vorticity, $\vec{\omega} = \nabla \times \vec{u}$, represents the curl of the velocity field $\vec{u}$, describing the local rotation of the fluid.

⁶Shear zones are characterized by a high gradient of velocity across a small region, often expressed mathematically as $\tau = \mu \frac{\partial u}{\partial y}$, where τ is the shear stress, μ is the dynamic viscosity, and $\frac{\partial u}{\partial y}$ is the velocity gradient in the direction normal to the flow.

2 On The Dataset

The dataset, derived from a `.lpk` file named `450mm_NoFlap_Seq_and_Avg.lpk` processed in `tecplot360`, contains data organized in a 3D matrix. It captures 801 timestamps, including 800 individual time instances and 1 average of all instances, each corresponding to 18,000 mesh points with dimensions $18,000 \times 17 \times 801$.

I had converted the `.lpk` into a `list` of 801 dataframes elements compressed into a `.pkl.gz` file format; this is done to make the amenable to `python` based processing.

Data Precision

- The measurements are precise up to six decimal points, ensuring high accuracy.

Data Dimensions and Structure

- **Matrix Dimensions:** Each timestamp comprises a data matrix for 18,000 mesh points across 17 columns.
- **Columns:** Data variables at each timestamp include:
 - Spatial coordinates: `X`, `Y` (in mm)
Here, `X` axis $\rightarrow$ Streamwise direction; `Y` axis $\rightarrow$ Spanwise direction & `Z` axis $\rightarrow$ Vertical direction.
 - Velocity components: `U`, `V` (in m/s)
 - Velocity gradients: `Ux`, `Uy`, `Vx`, `Vy`
 - Vorticity: `vor`
 - Mean velocities: `Umean`, `Vmean`
 - RMS velocities: `Urms`, `Vrms`
 - Velocity fluctuations: `ufluc`, `wfluc`
 - Additional variables: `V16`, `V17`
- **NOTE: Important Clarification** The dataset models a 2D flow with intended Z-axis disturbances:
 - `Y` actually represents `Z`
 - `V` represents `W`
 - `Uy`, `Vy` represents `Uz`, `Wz`
 - `Vmean`, `Vrms` actually represents `Wmean`, `Wrms`

Fig. 1a and **fig. 1b** gives a glimpse of the dataset for 451st and 801st timestamp respectively (**NOTE:** the timestamps have been indexed from 0 ie the first timestamp has index 0, the second timestamp has index 1 and so on).

Spatial Distribution

- **X Coordinates:** From 0.314097 to 88.957400 (in mm), 150 points are uniformly spaced.
- **Z Coordinates:** Intended to represent the Z-axis, spanning from -32.997800 to 33.032600 (in mm) with 120 uniformly spaced points.

Refer **fig. 2** (ie **2a** and **2b**) for the mesh-grids.

3 Method

The methodology employed in the study comprises of the following key techniques:

1. **Visualization:** Creation of contour plots for U_{mean} and W_{mean} , interpreting gradients and disturbances to infer turbulence presence.
2. **Turbulent Kinetic Energy (TKE) Analysis:** Calculation of TKE across the flow field to determine the turbulence intensity and its downstream evolution and carving relevant parameters, contour plots and animations.
3. **Turbulent Intensity (TI) Computation:** Determination of Turbulent Intensity using RMS of velocity fluctuations against mean and reference velocities to quantify turbulence strength.
4. **TNTI via TotalKE:** Identification of the TNTI by thresholding $Total_{KE}$, categorizing regions above specific thresholds as turbulent.

5. **Vorticity Thresholding:** Use of scalar vorticity and percentage-based thresholds to pinpoint turbulent regions for TNTI refinement.
6. **Critical w_{fluc} Analysis:** Identification of TNTI by establishing critical w_{fluc} thresholds through conditional statistics.

This multi-faceted approach allowed for a detailed characterization of the TNTI .

4 Results & Discussion

We have calculated various parameters and analysed them from the flow to make sense of the flow; and understand the dynamics of the flow. With a primary objective to measure the turbulence of the flow and to identify the TNTI.

4.1 Analysing U_{mean} and W_{mean}

Fig. 3 shows the contour of the mean velocity (U_{mean} and W_{mean}) for all the 18000 mesh points. Table 1 Contains few basic stats about the flow:

Table 1: Statistical Summary of Mean Velocities

	U_{mean}	W_{mean}
Mean	3.33	0.024
Standard Deviation	0.16	0.009
Minimum	2.92	-0.049
Maximum	3.76	0.063

Fig. 3a shows a smooth gradient of U_{mean} from low to high velocities as we move from the bottom of the plot (which represents the surface of the flat plate) upwards. This gradient is typical for boundary layer flow over a flat plate, where the velocity increases from zero at the wall (no-slip condition) to the free stream velocity away from the wall. The color gradient is relatively uniform along the length of the plate (X-direction), indicating steady flow with minimal variation in the streamwise direction (ie along the X-direction).

It is observed from fig. 3b that the W_{mean} is very small compared to corresponding U_{mean} ($\sim$ two order of magnitude smaller). Therefore we can interpret the W_{mean} as disturbances; which suggests a complex interplay of upward and downward moving fluid particles, possibly indicative of the turbulent nature of the flow or the influence of external perturbations .

Computing Turbulent Kinetic Energy

Fig. 4 illustrates the distribution of Turbulent Kinetic Energy (TKE) across a flow field above a flat plate.

$$TKE = \frac{1}{2}(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}) \quad (1)$$

In (1) where u' , v' , and w' are the fluctuating components of the velocity in the streamwise, vertical, and spanwise directions, respectively, and the overbar denote averaging, over the 800 timestamps. It quantifies the kinetic energy per unit mass contained in turbulent eddies.

In our data the TKE varies approximately from 0.070 to 0.279 m^2/s^2 . A higher TKE value corresponds to more intense turbulence and greater energy within the turbulent eddies. From fig. 4 one can infer that the TKE appears to increase in the streamwise direction. This could indicate that the turbulence intensity grows as the flow develops along the flat plate. Physically, this can be associated with the thickening of the turbulent boundary layer or the downstream evolution of flow instabilities. Further their is significant variation in TKE along the spanwise direction, highlighting the 3D nature of the turbulence. These suggests the presence of a fully developed turbulent flow regime over the plate, with sustained or increasing turbulence intensity in the downstream direction. This has implications for boundary layer theory, indicating a momentum transfer from the free stream to the wall, which impacts skin friction drag and potentially flow separation characteristics.

4.2 Computing Turbulent Intensity

Turbulent Intensity (TI) is a dimensionless quantity representing the ratio of the root-mean-square (RMS) of the velocity fluctuations to the reference velocity in a flow field. Mathematically it is defined as follows:

$$TI_{U_{mean}} = \frac{RMS(\vec{u})}{||\vec{U}_{mean}||} \quad (2a)$$

$$TI_{U_{ref}} = \frac{RMS(\vec{u})}{||\vec{U}_{ref}||} \quad (2b)$$

In (2a) and (2b) $\vec{u}_{rms}$ is the RMS of the fluctuating component of the velocity and $\vec{U}_{mean}$ is the Mean Velocity flow field & $\vec{U}_{ref}$ is the reference velocity (which in our case will be the free stream velocity $\vec{U}_{free\ stream}$).

Table 2: Statistical Summary of Turbulent Intensity

	$TI_{U_{mean}}(\text{in}\%)$	$TI_{U_{ref}}(\text{in}\%)$
Mean	12.42	12.40
Standard Deviation	1.76	1.67
Minimum	8.30	7.96
Maximum	16.04	15.85

From fig. 5a, one can observe that $TI_{U_{mean}}$ range from about 8% to over 15% (refer 2, signifying that the **turbulence in this flow regime is relatively intense**. High $TI_{U_{mean}}$ values can lead to enhanced mixing and increased momentum transfer. The plot further shows a change in $TI_{U_{mean}}$ across the span of the plate; suggesting the presence of spanwise inhomogeneities in the flow.

While plotting fig. 5b, we considered the $U_{ref} = \vec{U}_{free\ stream}$; $||\vec{U}_{free\ stream}|| \approx 3.34$ m/s. Further observe that the distribution of the two TIs are fairly similar (2 and 5). Note that in the previous plot 5a we have used the local mean velocity ($\vec{U}_{mean}$) as the reference, which would inherently adjust for variations in the flow's speed at different locations. The current plot 5b uses a constant reference velocity, highlighting the relative strength of turbulent fluctuations independently of the local mean flow speed.

A qualitative comparison between 5a and 5b suggests the following:

1. **Dominant Turbulent Fluctuation:** The velocity fluctuations are dominant over the mean flow field
2. **Mild Mean Velocity Variation:** Large variations in $\vec{U}_{mean}$ would likely result in noticeable differences between the two plots (fig. 5)

4.3 Computing Total Kinetic Energy

Total Kinetic Energy (Total_KE) encapsulates the entire kinetic energy contained within a fluid flow, considering both the ordered, macroscopic motion and the random, microscopic fluctuations due to turbulence.

$$\text{Total_KE} = \frac{1}{2} \vec{U} \cdot \vec{U} \quad (3)$$

In (3), $\vec{U} = \vec{U}_{mean} + \vec{U}_{fluc}$; where $\vec{U}_{mean} \rightarrow$ mean velocity & $\vec{U}_{fluc} \rightarrow$ fluctuating component of the velocity. Further observe that $\vec{U}_{mean} = (U_{mean}, V_{mean}, W_{mean})$ and $\vec{U}_{fluc} = (u', v', w')$. We will ignore the Y component of the velocity in our analysis. Refer fig. 6 for Total_KE contour for timestamp 1 and 53

Let us define two parameters **Ratio1**, $R1$ and **Ratio2**, $R2$ as follows:

$$R1 = \frac{\text{Total_KE} - \text{MeanFlow_KE}}{\text{MeanFlow_KE}} \quad (4a)$$

$$R2 = 1 + R1 \quad (4b)$$

Where $\text{MeanFlow_KE} = \frac{1}{2} \vec{U}_{mean} \cdot \vec{U}_{mean}$. $R1$ and $R2$ is supposed to capture the impact of the interaction of the fluctuating components with the mean flow field.

Observe that TKE and MeanFlow_KE are subsets of Total_KE and $\text{Total_KE} = \text{TKE} + \text{MeanFlow_KE} + \text{KE due to interaction among the fluctuating component and Mean Flow field}$.

Table 3: Variation of the **R1** & **R2** for the dataset

Parameter	Link to the tecplot Animation
R1	link R1
R2	link R2

Observe that the value of $R2$ oscillates fairly periodically in a range of 0.4 to 2.9 . A value of $R2 > 1 \implies$ there is strong dominance of the fluctuating component of the flow which can induce turbulence.

4.3.1 Constructing a KE based parametric TNTI boundary

Our principal target was to design a boundary between the Turbulent and Non-Turbulent flow. In that regard we are considering a kinetic energy based filtering of the flow into turbulent and non-turbulent. We will consider a reference $\text{Total_KE} = 20.720\text{m/s}$ (this is the global maximum Total_KE in the flow; let us call it KE_{ref}). Now we would consider a $\text{KE}_{\text{threshold}} = \alpha \text{KE}_{\text{ref}}$; where $\alpha \in [0, 1]$. However we would primarily be concerned with $\alpha \geq 0.5$; especially $\alpha = 0.4, 0.5, 0.6, 0.7, 0.8$

In fig. 7 we have marked the points which have $\text{Total_KE} \geq \text{Threshold}$ value for $\text{Timestamp} = 1$. This is done so as to find the areas with complex flow structures, such as turbulent eddies or vortices, which is a result of boundary layer separation and interaction with the flow surfaces. Those regions are likely sites of vigorous energy cascade, where larger eddies break down into smaller ones, transferring energy down to scales where it can be dissipated through viscosity.

We have followed the same procedure for $\text{Timestamps} = 1, 2, \dots, 800$ and generated films to identify the Separation layers.

Table 4: TNTI boundary at various Total_KE Thresholds

Threshold Level	Link to the .py Animation
80% of global maxima	Click here
70% of global maxima	Click here
60% of global Maxima	Click here
50% of global maxima	Click here
40% of global maxima	Click here

4.4 Analysis using the Vorticity ($\vec{\omega}$) & the Fluctuating components of Velocity (ufluc & wfluc)

The dataset has `vor` (vorticity) provided to us for all the 800 timestamps, using which we form the vorticity plots for all the timestamps and create an animation. Further observe that our flow is 2D so the vorticity will be a scalar quantity. [Link to the vorticity animation.](#)

High vorticity areas typically correspond to turbulent regions with a high degree of fluid rotation, while low vorticity regions are more laminar or non-turbulent. The distribution of vorticity indicates the coherence of vortical structures, which play a key role in the mixing and entrainment processes in turbulent flows.

By applying a threshold to the vorticity magnitude, we intended to identify the TNTI. The choice of threshold affects the identification of the TNTI. A higher threshold may lead to a more conservative estimation of the turbulent region, potentially missing out on weaker turbulent structures. A lower threshold could include regions of transitional or intermittent turbulence. Just like in the previous section we have decided a threshold based on percentage of the global maxima of the vorticity in the dataset which in our case is 4027.970s^{-1}

Table 5: TNTI boundary at various ω Thresholds

Threshold Level	Link to the .py Animation
5% of global maxima	Click here
10% of global maxima	Click here
15% of global Maxima	Click here
20% of global maxima	Click here

It has already been mentioned that the $\vec{\omega}$ is a scalar quantity and therefore one can easily observe (from [Link to the Velocity Vector animation](#)) that the changes in $\vec{\omega}$ could be ascribed to the `wfluc` (fluctuating component of Zcomponent of velocity vector)

Our goal then reduces to finding some kind of thresholds on `wfluc`. We choose to simultaneously analyze the relationship between `ufluc`, `wfluc` & `vor` for all the 800 timesteps at all 18000 points.

4.4.1 Computing Conditional Statistics For `ufluc`, `wfluc` & `vor`

Let us define a variable `vor_fluc` $\equiv \text{vor} - \text{vor_mean}$; where `vor_mean` is the mean vorticity (averaged over 800 timesteps for all 18000 points separately). Fig. 8 has the `vor_mean` plotted.

The root mean square (RMS) and mean square (MS) of a quantity u & w are defined as follows:

$$\text{RMS}(u) = \sqrt{\frac{1}{n} \sum u^2} \quad (5a)$$

$$\text{RMS}(w) = \sqrt{\frac{1}{n} \sum w^2} \quad (5b)$$

$$\text{MS}(u, w) = \frac{1}{n} \sum uw \quad (5c)$$

Here n is the number of elements.

Algorithm Employed

1. **ref. velocity** $U_0 = \text{mean}\{U_{\text{mean}} \text{ for } 18000 \text{ points}\} = 3.33294 \approx \text{Free Stream Velocity}$
2. Selecting $w_{fluc, \text{critical}} = U_0 \alpha$; where α in $[0, 0.01, 0.02, \dots, 0.99, 1.0]$
3. Select a point of your wish (Index in $[0, 18000]$)
4. $n = 0$
5. For α in $[0, 0.01, 0.02, \dots, 0.99, 1]$:
 - (a) $w_{fluc, \text{critical}} = U_0 \alpha$
 - (b) Find the w_{fluc} such that $|w_{fluc}| \geq w_{fluc, \text{critical}}$ (also keep their count; $n=n+1$)
 - i. Store the corresponding u_{fluc} and ω_{fluc}
6. Now compute $\text{RMS}(u_{fluc}), \text{RMS}(w_{fluc}), \text{MS}(u_{fluc} \cdot w_{fluc}), \text{RMS}(\omega_{fluc})$ for the above data.

After observing the variation of $\text{RMS}(\omega_{fluc})$ with w_{fluc} for a series of random selected mesh points (refer Fig. 9). From Fig. 9, it can be inferred that the $\text{RMS}(\omega_{fluc})$ drops when w_{fluc} is in a range of $\sim 15 - 25\%$ of the Free stream velocity (U_0). Therefore an optimal critical threshold for separating the TNTI must lie somewhere between this $15 - 25\%$ of U_0 .

5 Conclusion

We successfully identified the TNTI within boundary layer flows over a flat plate using comprehensive data analyses and robust framework. Through the analysis of mean velocities (U_{mean} and W_{mean}), the contours revealed a classic boundary layer profile and indicated the presence of a fully developed turbulent regime. The turbulent kinetic energy (TKE) calculations highlighted an increase in turbulence intensity downstream, with significant spanwise inhomogeneities, characteristic of complex 3D turbulent structures.

The Turbulent Intensity (TI) provided a dimensionless assessment of the flow's turbulence, suggesting intense turbulent activity with fluctuations dominating the mean flow. The comparison of TIs, using local mean velocity and a constant reference velocity, provided insights into the flow's relative strength and the role of localized flow speeds.

Total Kinetic Energy (Total_KE) was paramount to understand the interaction between fluctuating components and the mean flow. The delineation of turbulent regions via thresholding Total_KE led to the identification of the TNTI. The findings suggest that regions with Total_KE above 60% of the maximum were likely to exhibit turbulent characteristics.

Vorticity-based analyses further refined the TNTI identification process. By employing vorticity thresholds, the study pinpointed regions of heightened rotational fluid motion, indicative of turbulence. The critical w -fluctuation thresholds provided a clear methodology for distinguishing between turbulent and non-turbulent zones, with an optimal threshold range identified between 15-25% of the free stream velocity.

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Appendix

- [Link](#) to the Code base

Images

Time Stamp: 450

	X	Y	U	V	Ux	Uy	Vx	Vy	vor	Umean	Vmean	Urms	Vrms	ufluc	wfluc	V16	V17
0	0.314097	33.0326	4.01821	0.004016	77.1230	-290.5410	167.7010	70.8794	458.2420	0.0	0.0	0.0	0.0	0.286227	-0.025978	450.314	0.648098
1	0.942293	33.0326	3.97692	0.089590	-45.3214	-325.8810	114.2250	107.9010	440.1060	0.0	0.0	0.0	0.0	0.231426	0.056238	450.942	0.641439
2	1.535590	33.0326	3.97183	0.137989	-30.3218	-394.4860	25.7398	77.9519	420.2260	0.0	0.0	0.0	0.0	0.223404	0.106301	451.536	0.640618
3	2.128890	33.0326	3.99644	0.132559	-93.5015	-286.4860	-32.7784	94.9265	253.7080	0.0	0.0	0.0	0.0	0.250220	0.103145	452.129	0.644587
4	2.722180	33.0326	3.96207	0.039420	-85.1898	-236.5000	-66.8625	-40.7594	169.6370	0.0	0.0	0.0	0.0	0.215870	0.009154	452.722	0.639043
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
17995	86.549700	-32.9978	3.28384	0.228372	-83.5318	-125.1040	15.7840	-98.4824	140.8880	0.0	0.0	0.0	0.0	-0.002790	0.201551	536.550	0.529652
17996	87.142900	-32.9978	3.20573	0.204296	-41.4645	-58.6936	-25.3254	-20.5813	33.3682	0.0	0.0	0.0	0.0	-0.062686	0.182417	537.143	0.517053
17997	87.736100	-32.9978	3.23490	0.179907	101.2760	-79.9917	-49.7241	26.6432	30.2676	0.0	0.0	0.0	0.0	-0.028819	0.160382	537.736	0.521758
17998	88.329300	-32.9978	3.33053	0.144912	212.4940	-160.4380	-75.0522	63.7831	85.3862	0.0	0.0	0.0	0.0	0.076060	0.129136	538.329	0.537162
17999	88.957400	-32.9978	3.50348	0.082506	242.6660	-259.4920	-145.5180	115.1970	113.9730	0.0	0.0	0.0	0.0	0.252439	0.077605	538.957	0.565077

18000 rows x 17 columns

(a) Snapshot of Dataset at Index 450; the 451st timestamp

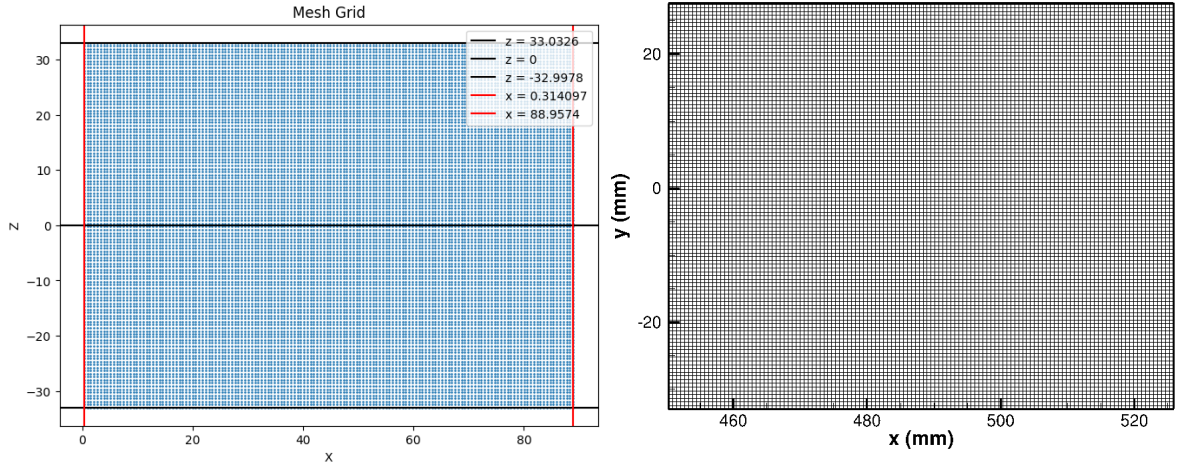
Time Stamp: 800

	X	Y	U	V	Ux	Uy	Vx	Vy	vor	Umean	Vmean	Urms	Vrms	ufluc	wfluc	V16	V17
0	0.314097	33.0326	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.73198	0.029995	0.537474	0.151676	-3.73198	-0.029995	450.314	0.0
1	0.942293	33.0326	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.74550	0.033352	0.531254	0.148314	-3.74550	-0.033352	450.942	0.0
2	1.535590	33.0326	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.74843	0.031688	0.532074	0.148811	-3.74843	-0.031688	451.536	0.0
3	2.128890	33.0326	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.74622	0.029415	0.536284	0.157544	-3.74622	-0.029415	452.129	0.0
4	2.722180	33.0326	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.74620	0.030266	0.540452	0.162781	-3.74620	-0.030266	452.722	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
17995	86.549700	-32.9978	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.28663	0.026822	0.654904	0.219418	-3.28663	-0.026822	536.550	0.0
17996	87.142900	-32.9978	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.26842	0.021879	0.646911	0.216320	-3.26842	-0.021879	537.143	0.0
17997	87.736100	-32.9978	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.26372	0.019525	0.647714	0.217231	-3.26372	-0.019525	537.736	0.0
17998	88.329300	-32.9978	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.25447	0.015776	0.650972	0.218226	-3.25447	-0.015776	538.329	0.0
17999	88.957400	-32.9978	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.25104	0.004901	0.674681	0.220429	-3.25104	-0.004901	538.957	0.0

18000 rows x 17 columns

(b) Snapshot of Dataset at Index 800; the 801st timestamp; which contains the average of all the 800 timestamps

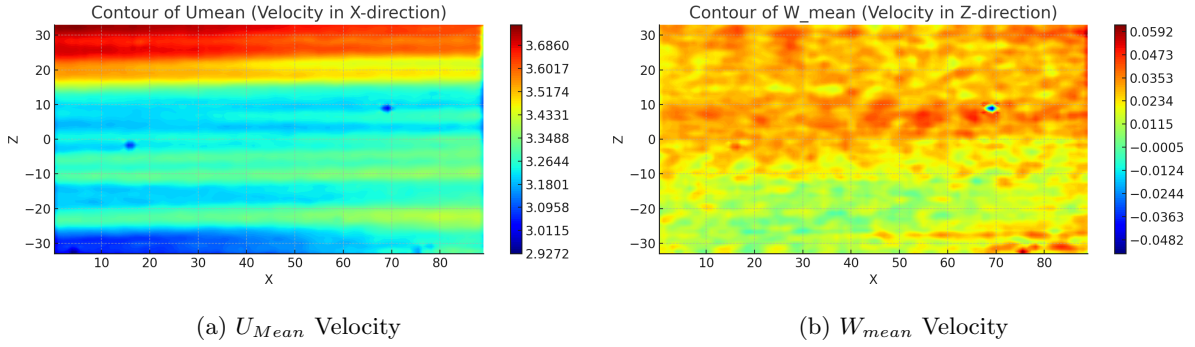
Figure 1: Snapshots of Dataset at Various Timestamps



(a) Mesh Grid generated using `.py` code

(b) Mesh Grid generated using `tecplot` (NOTE: `y` here is actually `z`)

Figure 2: Overall view of Mesh Grids



(a) U_{Mean} Velocity

(b) W_{mean} Velocity

Figure 3: Mean Velocity Contour for all the 18000 locations; averaged over the 800 timestamps

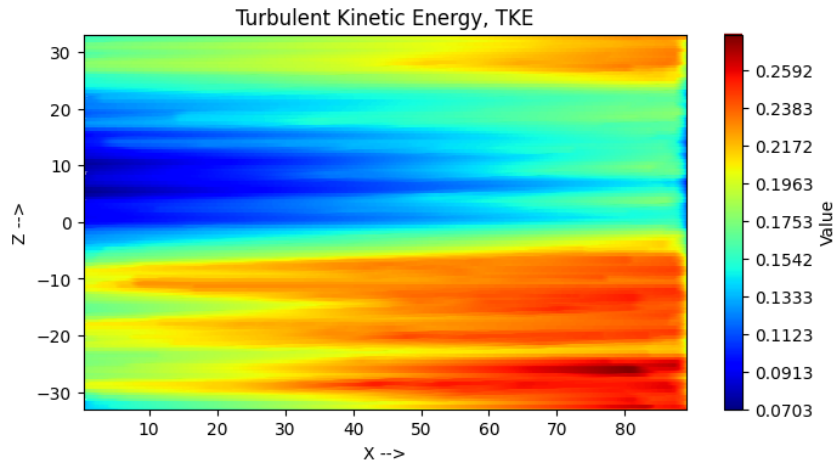
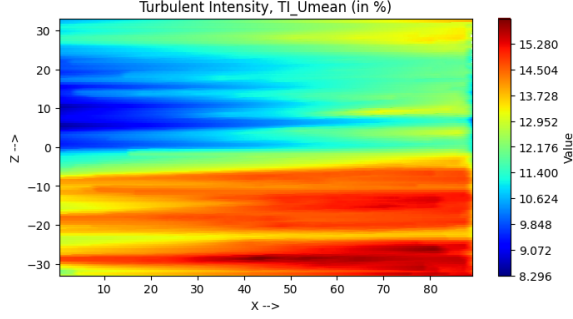
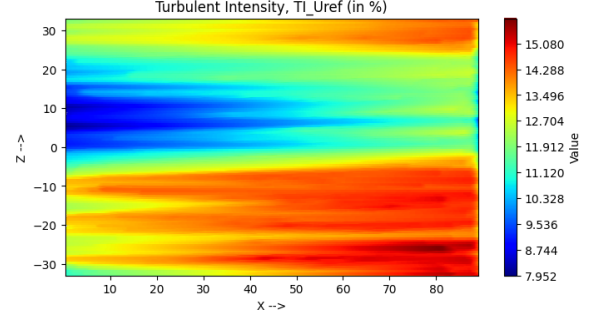


Figure 4: TKE for all the 18000 points

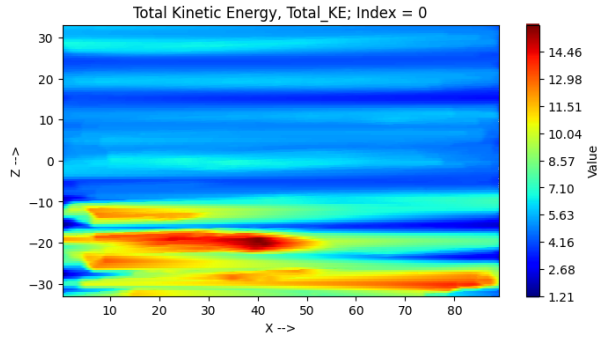


(a) TI based on local mean velocity

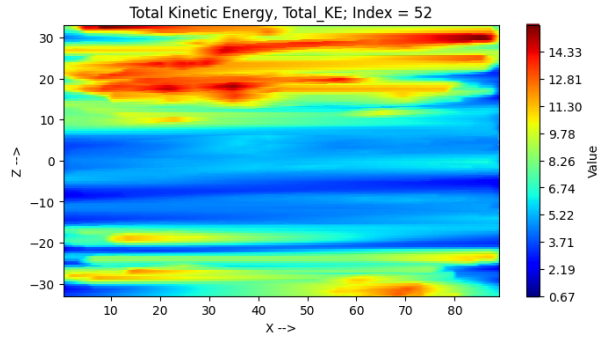


(b) TI based on reference velocity

Figure 5: Comparison of Turbulent Intensity distributions using different reference velocities



(a) Total KE at Timestamp = 1



(b) Total KE at Timestamp = 53

Figure 6: Total Kinetic Energy at for different timestamps

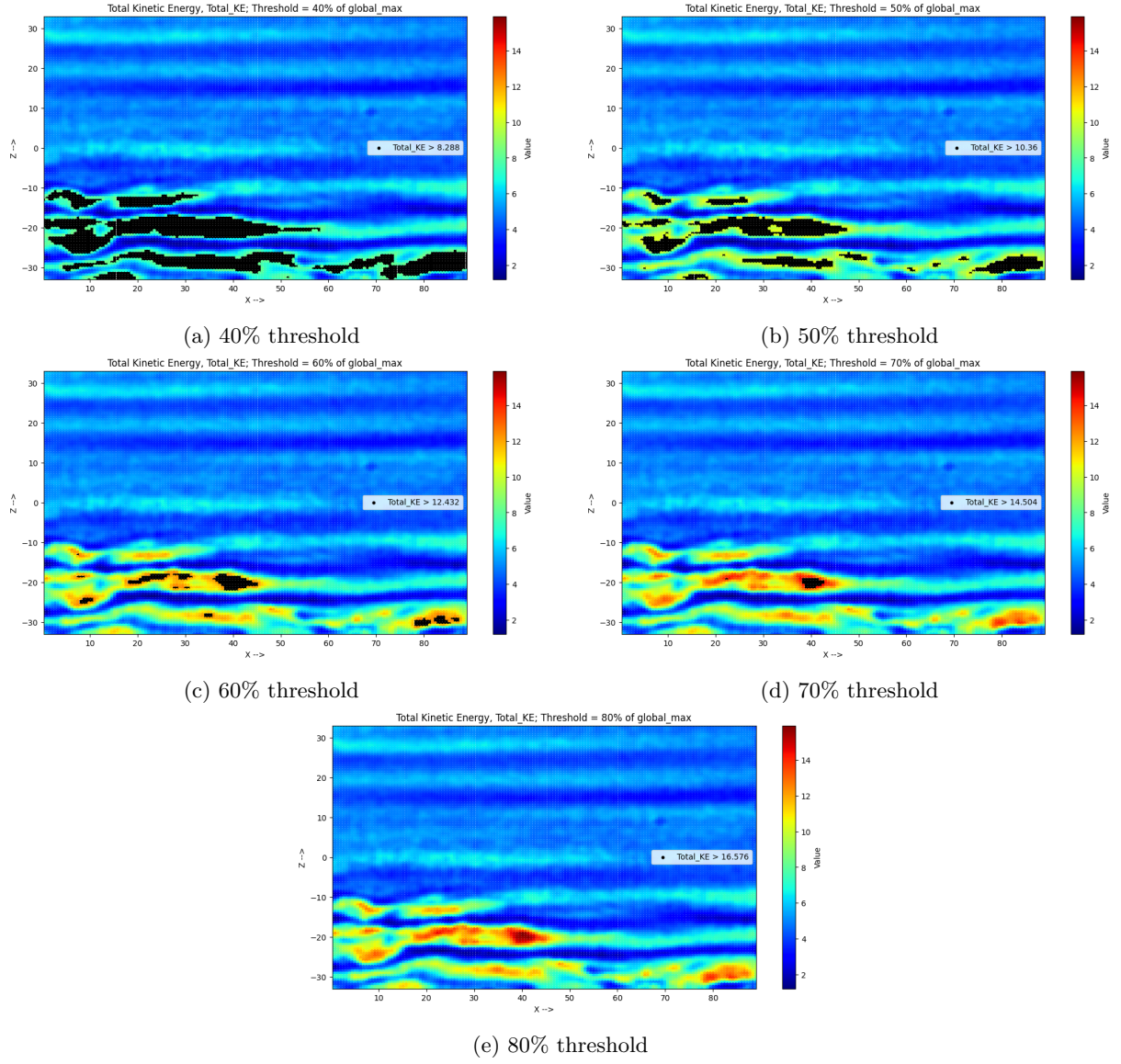


Figure 7: Identifying mesh points where the Total_KE exceeds a given threshold; Timestamp = 1

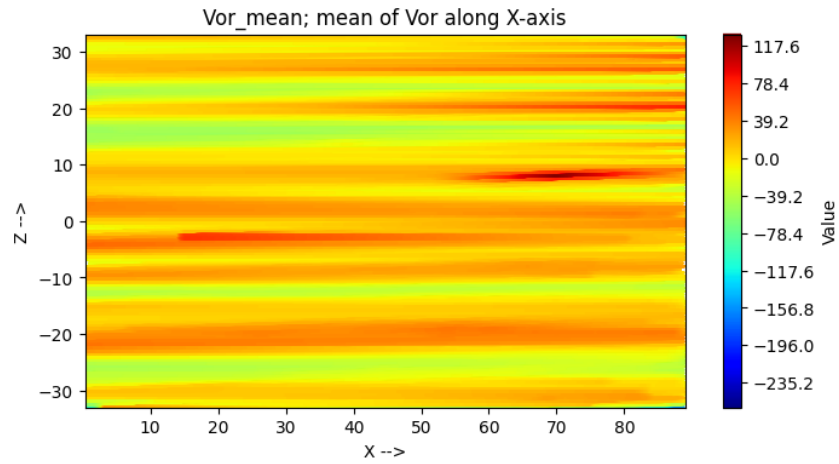
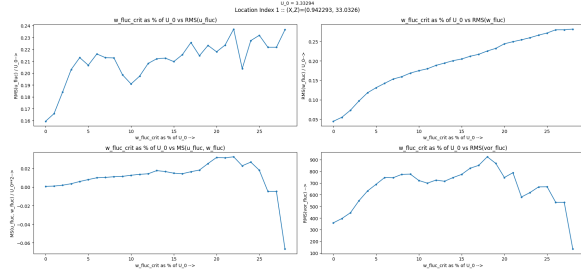
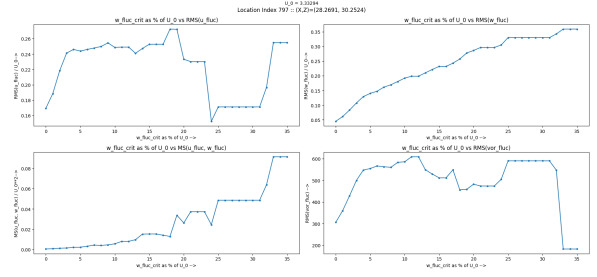


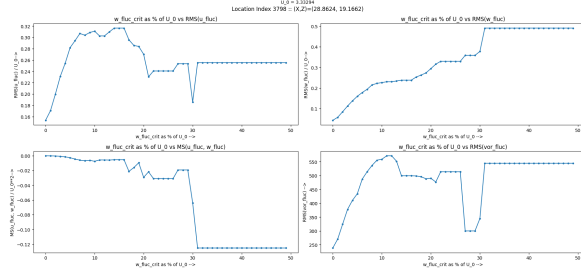
Figure 8: Vor_mean for all the 18000 points



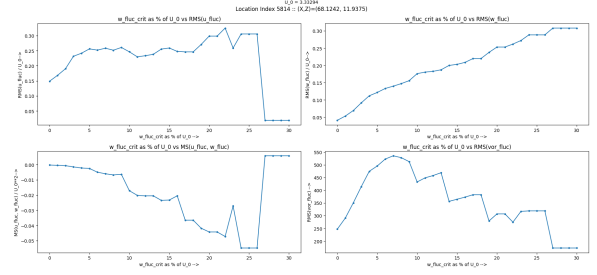
(a) Location Index = 1



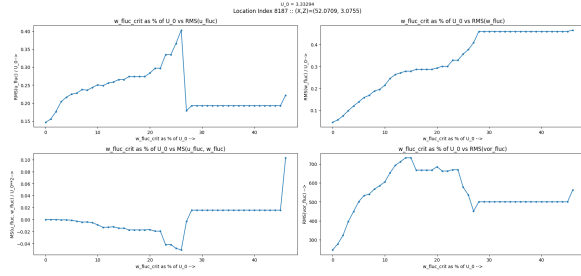
(b) Location Index = 797



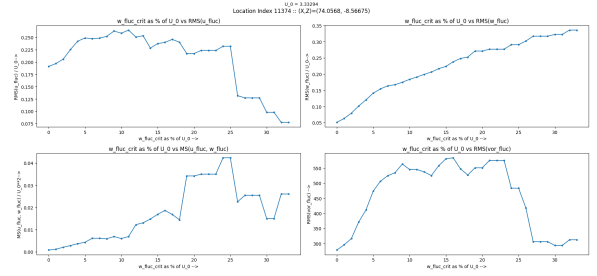
(c) Location Index = 3798



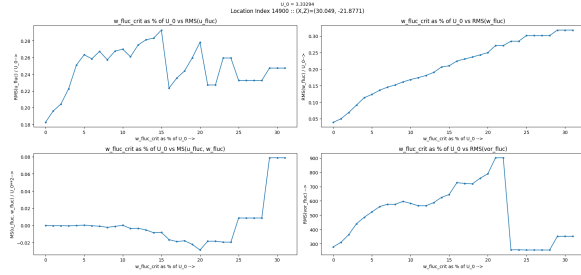
(d) Location Index = 5814



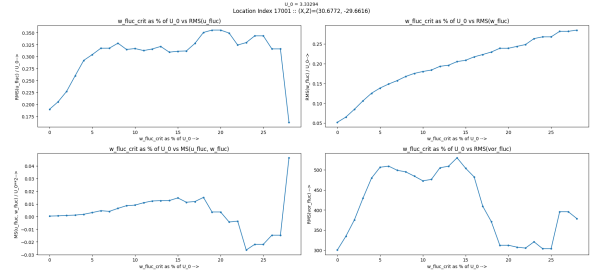
(e) Location Index = 8187



(f) Location Index = 11374



(g) Location Index = 14900



(h) Location Index = 17001

Figure 9: The RMS and MS plots for various $w_{fluc, crit}$ at different locations